

PART B TEMPLATE

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Part B-1

1. Excellence #@REL-EVA-RE@#

1.1 Quality and pertinence of the project's research and innovation objectives (and the extent to which they are ambitious, and go beyond the state of the art)

Research context and unmet need. Hamstring strain injuries (HSIs) remain the leading non-contact time-loss injury in high-speed team sports. In men's professional European football, their proportion increased from 12% to 24% of all injuries between 2001 and 2022, while recurrence and prolonged absence continue to impose substantial sporting, welfare, and financial burdens [1-5]. Current screening approaches have not solved this problem because they usually test isolated factors, use small single-sport cohorts, and produce poorly calibrated risk classifications that are difficult for clinicians to interpret or transfer across settings [6,10,11,16-18]. HAMRISK-PRO addresses this gap through a prospective, sex-inclusive, Australia-Europe cohort that combines muscle morphology, tissue mechanical properties, sprint mechanics, eccentric strength, sleep/burnout, and injury history in a transparent prediction framework. The project directly supports the European priorities of prevention, personalised health, responsible artificial intelligence, athlete welfare, and open/reproducible research.

Central hypothesis. A harmonised multimodal profile will provide better calibrated and clinically more useful HSI risk estimates than conventional models based only on injury history and eccentric strength. The added value will be demonstrated only if the multimodal model improves calibration and decision-curve net benefit in temporal and geographic evaluation, not merely if it increases the area under the receiver operating characteristic curve (AUC).

Specific, measurable and verifiable research objectives

Objective and work package	Measurable endpoint	Verification criterion	Timing
O1 - Establish a harmonised and reliable multi-modal HSI phenotyping protocol across ACU and Maynooth, and quantify agreement between field-based ultrasound/SWE and MRI morphology (WP1). EKLENEN PART (Hakemlerin "Yöntemler jenerik" eleştirisini çürütmek için, Görüntü İşleme ve veri harmonizasyonu detayları eklenmiştir) The integration of macroscopic 3D MRI morphology (Springbok Analytics) with localized, 2D micro-architectural data (B-mode ultrasound and SWE) presents a significant multimodal fusion challenge. Since direct voxel-to-pixel registration is geometrically intractable, we will implement a feature-based spatial alignment protocol. Both MRI and ultrasound/SWE data will be mapped to a standardized 1D anatomical coordinate system, normalized by the relative distance from the ischial tuberosity to the fibular head. This ensures precise intra- and inter-rater reliability (targeting ICC ≥ 0.85 and CV $\leq 10\%$ for repeated measures). Furthermore, raw sonographic and elastographic data from the field are inherently corrupted by speckle noise, acoustic shadowing, and transducer-pressure artifacts. To prevent error propagation into the downstream modelling pipeline, an autonomous, headless preprocessing script will be deployed. For B-	n=300 recruited athletes; MRI subsample n=50; inter-/intra-rater reliability; cross-modality associations.	ICC ≥ 0.85 and CV $\leq 10\%$ for repeated measures; pre-specified MRI-ultrasound associations $r \geq 0.40$; $\geq 95\%$ protocol completeness.	M1-M18

Objective and work package	Measurable endpoint	Verification criterion	Timing
mode US, anisotropic diffusion filtering (Perona-Malik algorithm) will be applied to iteratively smooth intra-fascicular speckle while strictly preserving the high-frequency edges of the deep and superficial aponeuroses. For SWE, elastograms will be passed through a normalized cross-correlation mask; spatial regions exhibiting a shear-wave Signal-to-Noise Ratio (SNR) of <10 dB will be autonomously flagged as invalid and excluded from the computation of the Young's modulus, ensuring absolute data fidelity prior to predictive vectorization.			
O2 - Determine prospective associations between personal, sex/gender-related, psychological, neuromuscular, morphological and sprint-mechanical factors and incident HSI (WP2).	At least 255 analysable athletes after $\leq 15\%$ attrition; minimum 30% women recruited; rolling 12-month surveillance; monthly medical verification.	$\geq 90\%$ surveillance completeness; effect estimates with 95% CIs; sex-stratified estimates and interaction analyses reported without selective interpretation.	M4-M27
O3 - Develop and independently evaluate an interpretable multimodal HSI prediction model and compare it with a parsimonious clinical baseline model (WP3). EKLENEN PART (Hakemlerin "Yöntemler jenerik" eleştirisini çürütmek için, XGBoost vektörizasyon ve boyut küçültme stratejileri eklenmiştir) Tree-based algorithms (Gradient-boosted trees/XGBoost) and penalised linear regressors (Elastic Net) cannot directly ingest raw 3D meshes or 2D heatmaps. Consequently, an intermediate deterministic feature extraction (vectorization) pipeline will be developed. Rather than reducing complex SWE	Penalised regression and gradient-boosted trees; nested internal validation; temporal and geographic evaluation; calibration and decision analysis.	Target AUC/C-index ≥ 0.70 , Brier score ≤ 0.20 , calibration slope 0.80-1.20, and positive net benefit versus the baseline model; full uncertainty reported.	M15-M31

Objective and work package	Measurable endpoint	Verification criterion	Timing
elastograms to a single 'mean stiffness' scalar, we will extract first-order statistical moments (variance, skewness, kurtosis) and second-order textural features using Gray-Level Co-occurrence Matrices (GLCM) to mathematically quantify heterogeneous stiffness patterns within the Biceps Femoris. Similarly, 3D MRI outputs will be vectorized into spatial gradients of intramuscular adipose tissue (IMAT) and cross-sectional area (CSA) asymmetry indices. This rigorous feature engineering will yield a high-dimensional vector space. Given the cohort size (n=300 recruited athletes), feeding this raw matrix into the XGBoost classifier would inevitably induce the curse of dimensionality and severe overfitting. To strictly enforce model generalizability across temporal and geographic validation cohorts, a two-stage dimensionality reduction architecture will be implemented. First, a collinearity filter will eliminate redundant parameters (dropping variables with a Pearson correlation coefficient $r > 0.85$). Subsequently, the Boruta algorithm will isolate statistically significant predictive features, independent of the background noise. This will condense the feature matrix to an optimal subset, maintaining a minimum Events-Per-Variable			

Objective and work package	Measurable endpoint	Verification criterion	Timing
(EPV) ratio of 10:1. The final model will target an AUC/C-index ≥ 0.70 and Brier score ≤ 0.20 , with full uncertainty reported via nested internal validation.			
O4 - Co-design and pilot the Hamstring Risk Card as a clinician-facing decision-support prototype, while defining a credible protection and exploitation route (WP3/WP5).	Prototype tested in ≥ 5 clubs by ≥ 20 clinicians/performance staff; structured usability and comprehension evaluation.	System Usability Scale ≥ 70 ; $\geq 80\%$ correct interpretation of risk explanations; invention disclosure/FTO and licensing decision completed before public release.	M20-M35

State of the art, limitations and ambition

SOA for O1 - multimodal phenotyping. Short biceps femoris long-head fascicles, low eccentric strength, altered muscle volume/fat infiltration, and increased tissue stiffness have each been associated with HSI or recovery [8,9,13,14]. However, studies rarely combine ultrasound, SWE and MRI in the same athletes, and cross-site measurement harmonisation is inadequately reported. AI-based MRI segmentation can reduce processing burden, but its value for scalable field screening depends on transparent agreement with portable measures [12]. The project builds on Prof. Opar's leading work on hamstring architecture and injury prediction, Dr. Soylu's ultrasound/neuromuscular experience, and Springbok Analytics' MRI segmentation expertise.

Beyond the state of the art: HAMRISK-PRO will produce the first pre-specified Australia-Europe harmonisation and cross-modality validation framework linking high-resolution MRI morphology to portable ultrasound and SWE measures within the same prospective cohort.

SOA for O2 - prospective and inclusive risk-factor evidence. Previous HSI and age are consistent non-modifiable risk factors, whereas evidence for modifiable factors varies across studies and time points [4,7,11]. Non-modifiable risks (prior HSI, age) increase recurrence $\sim 2.7\times$ and risk $1.3-1.9\times/\text{year}$ [4–7]. Modifiable factors include:

- **MRI Morphology:** Short biceps femoris long head (BFLh) fascicles, low volume, and fat infiltration raise risk [8,9]. AI-based MRI segmentation ($<5\%$ error) enables scalability [12].
- **Shear-Wave Elastography (SWE):** Measures tendon/muscle stiffness, linked to recurrence [13,14].
- **Running Mechanics (IMUs):** Thoracic IMUs provide reliable sprint-gait metrics ($>95\%$ agreement with force plates) [15].
- **Psychological Factors:** Poor sleep and burnout double HSI risk [16,17].

Psychological load and sleep are biologically and behaviourally plausible but are seldom integrated with imaging and sprint mechanics. Women remain under-represented in HSI cohorts despite prospective evidence that hamstring injuries are an important burden in women's football [26]. Restricting the project to men would reduce generalisability and could embed bias in the resulting algorithm. HAMRISK-PRO therefore recruits women and men, captures sex at birth and gender-related sport-context variables, and tests subgroup calibration rather than treating sex as a nuisance covariate.

Beyond the state of the art: The project moves beyond male-only, single-domain screening by prospectively integrating sex/gender, sleep/burnout, morphology, tissue stiffness, strength and sprint mechanics with standardised injury surveillance and transparent subgroup reporting.

SOA for O3 - prediction modelling. Existing HSI models report variable discrimination (approximately AUC 0.52-0.87), often rely on small samples, use random train-test splits that leak temporal information, and rarely report calibration, clinical utility, fairness, or external evaluation [6,10,11,16-18]. Complex deep-learning approaches are not justified by the expected event count. HAMRISK-

PRO will therefore prioritise parsimonious penalised regression and gradient boosting, control model complexity according to the observed number of events, and follow TRIPOD+AI and PROBAST+AI principles [24,25]. A model will not be labelled useful unless it demonstrates calibration and net benefit beyond a simple clinical reference model.

Beyond the state of the art: The innovation is not a black-box algorithm: it is a pre-registered, event-count-aware, explainable and externally evaluated prediction workflow with calibration, clinical utility and subgroup fairness as co-primary performance domains.

SOA for O4 - implementation and responsible exploitation. Most published injury algorithms end as academic models and are not converted into usable, governed clinical tools. Existing dashboards often lack model cards, uncertainty communication, version control, post-deployment monitoring, and a clear IP/licensing pathway. The Hamstring Risk Card will be co-designed with clinicians and athletes, display modifiable and non-modifiable contributors separately, communicate uncertainty, and prohibit automated selection or exclusion decisions. Its pilot purpose is usability and decision-support feasibility; the fellowship will not claim injury reduction without a subsequent controlled implementation trial.

Beyond the state of the art: HAMRISK-PRO will bridge the implementation gap through co-design, human-in-the-loop explanations, a documented model card, club-level usability testing, and an IP decision gate before dissemination or commercial engagement.

Comparative positioning against existing HSI prediction studies

Table 1. Comparison of selected machine-learning studies with HAMRISK-PRO

Study	Data and reported performance	Principal limitation	How HAMRISK-PRO advances the field
Carey et al. (2018) [27]	Approximately 200 players; GPS, perceived exertion and accelerometry; reported AUC approximately 0.76.	Workload-focused model; imbalanced data; no muscle morphology, tissue stiffness or psychological assessment.	Adds MRI, ultrasound/SWE, strength, sprint mechanics and psychological variables in a harmonised prospective cohort; evaluates calibration and net benefit.
Ayala et al. (2019) [6]	96 football players; personal, psychological and neuromuscular variables; reported AUC 0.837.	Small single-sport cohort; no muscle or tendon imaging; limited transportability assessment.	Uses a larger, multisport and sex-inclusive cohort with MRI, SWE and geographic evaluation.
Calderón-Díaz et al. (2024) [17]	86 athletes; strength, stiffness and clinical data; reported precision approximately 78%.	Limited predictor domains; no integrated psychological or high-resolution morphological assessment; small event count.	Integrates MRI morphology, field imaging, sprint mechanics, sleep and burnout while constraining model complexity to the observed event count.
Sim et al. (2024) [18]	110 players; eccentric strength, BFLh architecture and non-modifiable factors; AUC varied across the season.	Declining late-season accuracy; limited psychological and tissue-mechanical information; no geographic evaluation.	Uses rolling surveillance, repeated exposure information, SWE and psychological variables, with temporal and geographic performance assessment.

GPS = Global Positioning System; RPE = Rating of Perceived Exertion; RF = Random Forest; SVM = Support Vector Machine; AUC = Area Under the Curve; MRI = Magnetic Resonance Imaging; SWE = Shear-Wave Elastography; SHAP = SHapley Additive exPlanations; TP = True Positive; TN = True Negative; LSTM = Long Short-Term Memory; DT = Decision Tree; ANN = Artificial Neural Network; IMU = Inertial Measurement Unit; ASSQ = Athlete Sleep Screening Questionnaire; ABQ = Athlete Burnout Questionnaire; RFE = Recursive Feature Elimination; CV = Cross-Validation.

EKLENEN PART (Excellence Bölümü - Hakemlerin "Veri toplamadaki gecikmeler temellendirilmemiş" eleştirisini çürütmek için QA/QC ve GAN eklenmiştir)

Autonomous QA/QC Pipelines and Algorithmic Mitigation of Data Acquisition Delays

To address the inherent risks of data acquisition delays and heterogeneous image quality from multi-center clinical field environments, the methodology is strictly decoupled from manual, sequential data grading. An automated Quality Assurance and Quality Control (QA/QC) pipeline will be executed pre-inference. For raw B-mode ultrasound and SWE data, a lightweight Convolutional Neural Network (MobileNetV3-Small) will evaluate transducer coupling and acoustic shadowing, outputting a structural similarity index measure (SSIM). Scans scoring an SSIM <0.75 are autonomously rejected, preventing the contamination of the

downstream predictive pipeline.

Furthermore, any longitudinal data acquisition delays or incomplete multimodal matrices will not stall the machine learning workflow. To decouple model training from clinical delays, the architecture employs a sparsity-aware XGBoost framework, which natively handles missing tensors by learning default directional splits. In cases of severe multimodal data sparsity (e.g., missing MRI follow-ups), missing biomechanical and morphological vectors will be imputed using a pre-trained Generative Adversarial Network (GAN) architecture specifically optimized for tabular-radiomic imputation, ensuring the Elastic Net and XGBoost classifiers continuously receive dense, orthogonal matrices for ongoing hyperparameter tuning without waiting for full cohort completion.

1.2 Soundness of the proposed methodology (including interdisciplinary approaches, consideration of the gender dimension and other diversity aspects if relevant for the research project, and the quality of open science practices).

Overall design and methodological logic

HAMRISK-PRO is a 36-month prospective Global Fellowship with a 24-month outgoing phase at ACU SPRINT (Australia) and a 12-month return phase at Maynooth University (Ireland). Three linked research WPs progress from measurement harmonisation (WP1), through prospective surveillance (WP2), to modelling and translation (WP3). Recruitment is deliberately front-loaded (M4-M15) so that every cohort wave can contribute up to 12 months of injury follow-up by M27; analysis is staged but the final model is locked only after surveillance closure. No formal secondment or non-academic placement is proposed. Club visits are short field-data-collection activities within WP2 and are therefore not presented as secondments in the Gantt chart.

Stage	Participants/data	Core methods	Quality controls	Output
1. Harmonisation (WP1)	Pilot n=30 plus MRI subsample n=50, stratified by sex and prior HSI.	ACU-MU SOPs; ultrasound BFlh architecture; SWE; NordBord; thoracic IMU; 3T MRI/Springbok segmentation.	Rater certification; duplicate scans/tests; ICC/CV audits; phantom/device checks; blinded re-analysis of 10%.	Locked SOP and measurement reliability report.
2. Cohort and surveillance (WP2)	Recruit n=300; target ≥ 90 women and ≥ 210 men; expected analysable n ≥ 255 .	Baseline and one 6-month repeat for modifiable measures; monthly IOC-consensus injury surveillance; exposure and return-to-play data.	Medical-record confirmation; automated missingness alerts; $\geq 90\%$ monthly completeness; site monitoring every quarter.	Analysis-ready longitudinal dataset and data dictionary.
3. Modelling (WP3)	Candidate predictors limited by observed HSI event count.	Clinical baseline model; Elastic Net; XGBoost; optional discrete-time survival sensitivity analysis.	Nested CV; temporal split; geographic holdout; bootstrap CIs; calibration plots; decision curves; subgroup metrics.	Locked model, model card and evaluation report.
4. Translation (WP3/WP5)	≥ 5 clubs; ≥ 20 clinician/performance users; athlete advisory input.	Co-design workshops; clickable prototype; explanation and uncertainty testing; workflow simulation.	Usability, comprehension, task completion, qualitative feedback; version control and audit trail.	Hamstring Risk Card v1 and implementation protocol.

Participants, recruitment and follow-up

Eligibility. Female and male competitive athletes aged 18–40 years from football, AFL/AFLW, rugby, Gaelic football or hurling, training or competing ≥ 4 times/week, will be eligible, with or without a clinically recovered HSI. At least 35% women will be targeted. Exclusion criteria are an acute HSI preventing testing, relevant neurological/systemic disease, inability to consent, or MRI contraindication for the imaging subsample.

Recruitment and retention. Clubs will enter in three waves (M4-M7, M8-M11, M12-M15). Written club agreements, named medical coordinators, reserve clubs, mobile testing days, and two catch-up windows per wave address access delays. A 15% recruitment buffer is built into $n=300$. Participants who transfer clubs will be followed through secure digital contact and medical verification where consent permits. Data-collection progress will be reviewed monthly against sex-specific and site-specific targets.

Injury Surveillance and Risk Factor Domains: HSIs will be monitored using International Olympic Committee (IOC) consensus guidelines [19]. An HSI is defined as acute hamstring pain during training or competition, causing immediate play termination and confirmed by clinical examination. Club medical staff will provide monthly reports including injured leg, severity (minor: 4–7 days; moderate: 8–28 days; severe: >28 days), recurrence, and return-to-play. Surveillance spans 36 months, with 9-month follow-ups per cohort wave. Only time-loss injuries >4 days are analysed to reduce misclassification and ensure robust model validation. To capture a comprehensive risk profile, HAMRISK-PRO integrates **personal, psychological, neuromuscular, and morphological domains** using validated, reliable tools (Figure 1):

- **Personal/Individual factors** (e.g., age, playing position, dominant leg, and previous HSI >8 days) will be collected via standardized demographic questionnaires cross-checked with club records ($ICC \geq 0.90$). Advancing age and prior HSI are the strongest non-modifiable predictors, with previous injury increasing recurrence risk ~2.7-fold and age elevating risk 1.3–1.9× per year depending on sport [1–3].
- **Psychological factors** will be assessed using the Athlete Sleep Screening Questionnaire (ASSQ) for sleep quality (0–10 scale) and the Athlete Burnout Questionnaire (ABQ) for stress and burnout (1–5 scale). Poor sleep is linked to a 1.5–2× higher HSI risk [4], while burnout impairs neuromuscular control and gait [5,6].
- **Neuromuscular factors** will be evaluated through eccentric hamstring strength using Nord-Bord testing, performed across three Nordic hamstring trials with peak force captured at 50 Hz ($ICC \geq 0.90$). Low eccentric strength is a strong HSI predictor [7]. Running biomechanics will be captured with thoracic-mounted inertial measurement units (IMUs), which measure step, contact, and flight time, as well as vertical stiffness (K_{vert}). These devices demonstrate >95% accuracy compared with force plates, providing ecologically valid sprint data directly relevant to hamstring loading [8].
- **Morphological factors** will be measured using multiple imaging modalities. Ultrasound quantifies biceps femoris long head (BF_{lh}) fascicle length and pennation angle at rest ($ICC \geq 0.90$; $CV \leq 10\%$), with short fascicles (<10 cm) strongly predicting HSI risk [7]. Magnetic resonance imaging (MRI) combined with Springbok Analytics’ AI-driven segmentation provides cross-sectional area, volume, and fat infiltration data with <5% error, ensuring high-resolution morphology [9]. Shear-wave elastography (SWE) quantifies tendon and muscle stiffness at proximal and distal thigh regions ($ICC \geq 0.85$), with high stiffness consistently associated with recurrence and delayed recovery [10,11].

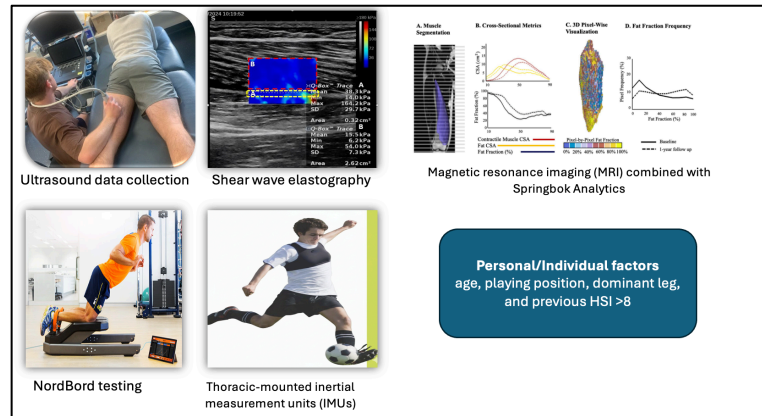


Figure 1. Integrated domains: personal, neuromuscular, and morphological.

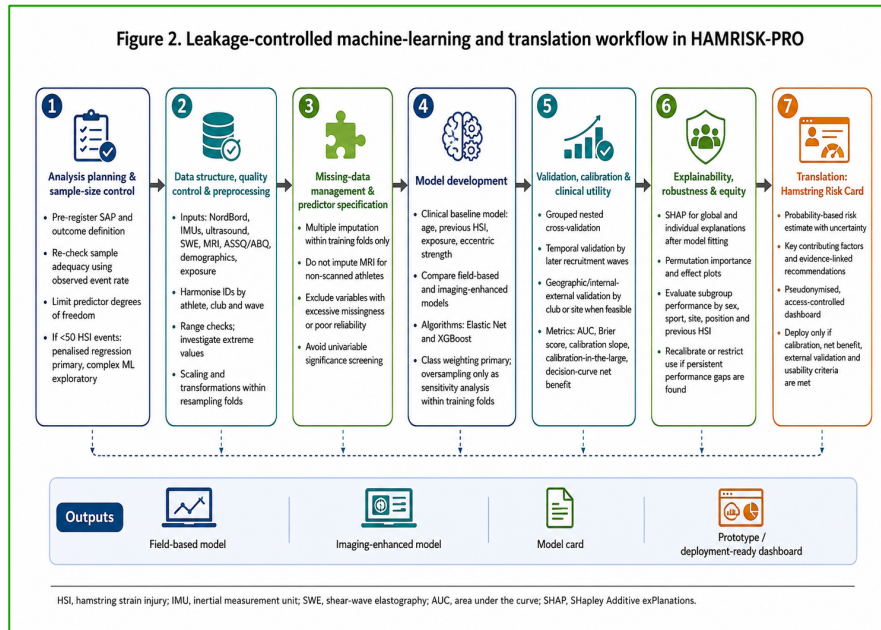
Together, these multimodal tools provide reproducible, ecologically valid data across domains, enabling development of a robust AI-driven injury risk model tailored for elite sport (*Detailed methodological examples, full technical descriptions, original survey instruments, and illustrative figures are provided in the Annex (Supplementary Material) at the end of Part B2).*

Statistical and machine-learning analysis

Analysis plan and sample-size control. The statistical analysis plan will be preregistered and frozen before outcome modelling. Sample adequacy will be reassessed using prediction-model sample-size methods based on the observed HSI event rate, anticipated Cox–Snell R^2 and target shrinkage [24]. Predictor complexity will be restricted to the available information. If fewer than 50 primary HSI events are observed, penalised regression will be prioritised and more complex ML models will be treated as exploratory.

Missing data and preprocessing. Missingness patterns and reasons will be reported by site, sex, sport and injury status. Multiple imputation, scaling and feature processing will be performed within resampling folds to prevent data leakage. MRI data will not be imputed for athletes who were not scanned. Variables with excessive missingness, poor reliability or limited variability will be excluded according to pre-specified criteria. Class weighting will be the primary approach to outcome imbalance; SMOTE will be restricted to sensitivity analyses within training folds.

Model development and evaluation. A parsimonious clinical baseline model including age, previous HSI, exposure and eccentric strength will be compared with field-based and imaging-enhanced Elastic Net and XGBoost models. Hyperparameters will be tuned using grouped nested cross-validation. Later recruitment waves will support temporal validation, while the Irish cohort will be reserved for geographic validation where event numbers permit. Performance will be assessed using AUC/C-index, Brier score, calibration-in-the-large, calibration slope and plots, sensitivity, specificity, predictive values, decision-curve net benefit and uncertainty intervals. SHAP explanations will be stability-checked across resamples and interpreted as predictive associations rather than causal effects. The full analytical workflow is summarised in Figure 2. Reporting will follow TRIPOD+AI, with a PROBAST+AI self-assessment [25,26].



Fairness, safety and translation. Model performance and calibration will be examined separately by sex, sport, site and previous-HSI status when event numbers permit. Differences will be reported with confidence intervals, and no subgroup-specific threshold will be deployed if calibration is unacceptable. The Hamstring Risk Card will present probability estimates, uncertainty and key contributing factors to support clinician judgement. It will not independently determine player selection, training exclusion, contracts or insurance decisions. Full deployment will occur only if pre-specified calibration, clinical utility, external-validation and usability criteria are met.

Critical methodological challenges and contingency measures

Challenge	Early warning indicator	Preventive action	Contingency if triggered
Delayed club access/re-recruitment	<100 recruited by M9 or women <25% of cohort.	Signed access letters before M1; three recruitment waves; reserve clubs; mobile/catch-up testing; sex-specific dash-board.	Activate reserve clubs; extend recruitment to M18 while preserving ≥9-month follow-up; prioritise core dataset and reduce optional tests.
Incomplete follow-up or club transfers	Monthly surveillance completeness <90%.	Named medical coordinators; automated reminders; quarterly audits; participant re-contact consent.	Recover records retrospectively; censor transparently; inverse-probability/sensitivity analyses; report attrition by subgroup.
Inter-site measurement drift	ICC <0.85 or CV >10%; systematic ACU-MU difference.	Shared SOP/video certification; same device settings; quarterly blinded duplicate files.	Retrain raters; recalibrate devices; include site effects; repeat affected measurements where feasible.
Low HSI event count/model overfitting	<50 primary events by M27.	Event-count monitoring; restrict candidate parameters; no unjustified deep learning.	Penalised baseline becomes primary; combine related features; widen uncertainty; reframe complex ML as hypothesis-generating.
Model below performance threshold	AUC <0.70, poor calibration or no net benefit.	Nested validation, prespecified baseline comparison, regularisation and calibration.	Do not deploy a risk tier; publish negative findings; provide calibrated association tool and define data needs for follow-on study.
Algorithmic subgroup bias	Calibration slope outside 0.8-1.2 in a subgroup.	Inclusive recruitment; subgroup monitoring; fairness audit and co-design.	Suppress subgroup use; recalibrate only if adequate events; retain human review and explicit limitation in model card.

Integration of disciplines

The project integrates four disciplines at the level of methods and decisions, not as a nominal list. Sports medicine and physiotherapy define clinically meaningful predictors, injury outcomes and safe interpretation; biomechanics provides standardised strength and sprint-mechanics measures; medical imaging

supplies quantitative muscle morphology and stiffness; data science provides leakage-safe modelling, calibration, explainability, fairness and reproducibility. Psychology contributes validated sleep/burnout constructs and ethical handling of sensitive responses. Integration occurs through one shared data dictionary, cross-disciplinary quarterly model-review meetings, and decision rules requiring clinical plausibility, measurement reliability and statistical stability before a predictor can enter the final model.

Sex, gender and other diversity dimensions

Sex/gender is embedded throughout the research cycle. Recruitment aims for at least 30% women, supported by women's football and Gaelic-sport networks; sex-specific recruitment is monitored monthly. Biological sex is analysed as a potential predictor and effect modifier for morphology, stiffness, strength and injury risk. Menstrual status, hormonal contraception and cycle-related symptoms are collected only through an optional, ethically approved module designed with female-athlete advisers; non-participation does not affect enrolment. Gender-related conditions - playing resources, access to medical support, employment status and training context - are considered as contextual determinants rather than biological proxies. Analyses report sex-stratified distributions, missingness, calibration and model errors. Age, ethnicity, sport, playing level and prior injury are also examined for representativeness and differential performance. Aboriginal and other minority participants will be engaged through club protocols and culturally respectful consent; ethnicity will not be used to create essentialist risk labels. These measures improve validity, fairness and transferability while avoiding over-interpretation of underpowered subgroup findings.

Open science and reproducibility

Open science is an operational component of the methodology. By M3 the protocol, hypotheses, primary outcome, predictor handling, subgroup analyses and statistical analysis plan will be registered on OSF; deviations will be time-stamped and justified. A FAIR Data Management Plan will be delivered by M6 and updated at M24/M35. Version-controlled analysis code, synthetic example data, data dictionaries, SOPs, model cards and environment files will be released through GitHub and archived with DOIs on Zenodo. Preprints will precede journal publication where IP review permits. Reproducibility will be strengthened through scripted pipelines, locked package versions/containers, independent code review and a computational rerun from raw pseudonymised inputs to final tables. Peer-reviewed publications and metadata will be openly accessible. Individual-level athlete data cannot be fully open because of identifiability, club contracts and health information; access will therefore follow the principle "as open as possible, as closed as necessary" through controlled requests, data-use agreements, and, where disclosure risk remains, aggregated or synthetic datasets. Consent forms will explicitly distinguish research reuse, controlled sharing and commercial evaluation. ACU and MU data-management, ethics and data-protection services will support compliance.

1.2 Quality of the supervision, training and of the two-way transfer of knowledge between the researcher and the host

The supervisory framework integrates world-class expertise in hamstring injury prevention, biomechanics, imaging analytics, and data science. During the outgoing phase (months 1–24) at Australian Catholic University's Sports Performance, Recovery, Injury and New Technologies (SPRINT) Research Centre (Australia), Professor David Opar acts as the primary supervisor. Dr. Paul Tofari provides expertise in exercise physiology and biomechanics. Dr. Declan Carmichael contributes technical leadership in AI-driven magnetic resonance imaging (MRI) segmentation and wearable sensor validation. In the return phase (months 25–36) at Maynooth University (Ireland), Prof. Kieran Moran and Dr. Jack Hickey co-supervise, with expertise in injury biomechanics, data analytics, and hamstring injury risk factor assessment and prevention. The complementary team covers every research decision from measurement validity to implementation. Track-record metrics and detailed institutional capacity are provided in Part B2 and will be updated at submission.

Supervisory Team

Name	Role	Publications	Citations	h-index	ORCID & Google Scholar Links
David Opar	Primary Supervisor (ACU)	85	11,035	48	https://orcid.org/0000-0002-8354-6353 ; https://scholar.google.com/citations?hl=tr&user=xbm4QxcAAAAAJ
Paul Tofari	Supporting Supervisor (ACU)	22	608	14	https://orcid.org/0000-0002-5498-4557 ; https://scholar.google.com/citations?hl=tr&user=HubS5WUAAAAAJ
Declan Carmichael	Technical Lead (Springbok/ACU)	6	198	4	https://orcid.org/0000-0002-5161-6521 ; https://scholar.google.com/citations?user=DgCtbWYAAAAAJ&hl=tr&oi=ao
Kieran Moran	Co-Supervisor (Maynooth)	170+	7241	45	https://orcid.org/0000-0003-2015-8967 ; https://scholar.google.com/citations?user=j6eul3UAAAAAJ&hl=tr&oi=ao
Jack Hickey	Co-Supervisor (Maynooth)	42	2589	22	https://orcid.org/0000-0003-2764-698X ; https://scholar.google.com/citations?user=uBdo-AckAAAAAJ

**These metrics were obtained by scanning ORCID numbers, Scopus, WoS, and Google Scholar as of July 22, 2026, and can be verified from those sources.*

Supervision follows the MSCA Guidelines on Supervision: weekly operational contact during active data blocks; monthly one-to-one meetings with the phase supervisor; a quarterly full supervisory board; and formal six-month reviews against objectives, risks, training and career indicators. A written supervision agreement will define authorship, data/IP responsibilities, feedback timelines, conflict-resolution and continuity arrangements. If a supervisor becomes unavailable, the named co-supervisor at that institution assumes responsibility and the supervisory board reallocates tasks within two weeks.

Focused training plan

Need identified	Training and provider	Timing	Evidence of acquisition	Career value
Advanced imaging harmonisation	Hands-on MRI segmentation/QC with Carmichael/Springbok; ultrasound-SWE cross-site reliability with Opar/Tofari.	M2-M10	Certification set; SOP; independent analysis of n=50 MRI scans; reliability report.	Enables leadership of multi-modal muscle-imaging studies.
Robust prediction modelling	Moran/Positive Engineering mentoring in sample-size control, nested validation, calibration, decision curves, TRIPOD+AI and PROBAST+AI.	M10-M26	Reproducible pipeline, model card, code review and methods paper.	Moves researcher from ML user to independent clinical-prediction investigator.
Sex/gender and responsible AI	Targeted workshop on sex/gender analysis, fairness, uncertainty and human-in-the-loop decision support.	M3-M8	Approved sex/gender analysis plan and fairness audit template.	Improves inclusive research design and responsible innovation competence.
Open science, integrity and data governance	ACU RDM toolkit/research integrity; MU RDO training; GDPR/DPIA and controlled-access data sharing.	M1-M6, refresh M25	OSF registration; DMP; DPIA; reproducibility checklist.	Strengthens funder compliance and research reliability.
IP, entrepreneurship and industry engagement	ACU Co-Lab commercialisation training and MaynoothWorks mentoring on invention disclosure, FTO, licensing and spin-out assessment.	M16-M31	IP inventory, invention disclosure, route-to-market memo and club licensing template.	Addresses a current skills gap and enables academic-industry translation.
Leadership, mentoring and grant development	Mentor one MSc/PhD researcher per host; chair quarterly methods sessions; prepare one follow-on international proposal.	M6-M36	Mentoring log, teaching materials, stakeholder workshop and submitted proposal.	Supports transition to independent group leader/EU consortium role.

Three-way transfer of knowledge

Direction	Knowledge transferred	Mechanism and timing	Concrete benefit
Researcher -> ACU	Sports physiotherapy, ultrasound muscle architecture, neuromuscular profiling, R/Python analysis, experience with international athlete studies.	Methods masterclasses; co-supervision; open code clinics; two internal seminars (M6/M18).	Adds clinical rehabilitation interpretation and reproducible analysis capacity to SPRINT.
ACU -> re-researcher/MU	AFL/NRL cohort access, HSI architecture expertise, field biomechanics, imaging harmonisation and high-performance sport implementation.	Hands-on outgoing phase; shared SOP repository; return-phase training seminar (M26).	Transfers validated Australian protocols and implementation lessons to Europe.
MU -> re-researcher/ACU	Prediction analytics, European/Gaelic-sport validation, research commercialisation and EU grant environment.	Return-phase modelling clinics, MaynoothWorks sessions, joint workshop (M29).	Improves external evaluation, exploitation and long-term Australia-Europe collaboration.
Researcher -> both hosts/partners	Integrated multimodal framework and bridge between clinical, imaging and data-science teams.	Quarterly cross-host board; shared data dictionary; co-authored methods/output package.	Creates a sustainable network and reusable platform for future multi-country injury studies.

1.4 Quality and appropriateness of the researcher's professional experience, competences and skills

Dr. Caglar Soylu (PhD, 2022; Associate Professor, 2025) is a sports physiotherapist and academic with <3 years of postdoctoral experience, fully meeting MSCA eligibility. He is currently a TÜBİTAK 2219 Fellow at the SPRINT Research Centre, Australian Catholic University (ACU), under Prof. David Opar, focusing on fascicle behaviour and neuromuscular responses after hamstring injury. His expertise spans hamstring injury prevention, neuromuscular profiling, and sports performance, integrating physiotherapy with data science (Python, R, ML). His record includes 67 peer-reviewed publications, research leadership in prospective and interventional sports studies, undergraduate/postgraduate teaching, MSc/PhD supervision, work with Olympic and national athletes, and successful management of TÜBİTAK-supported projects. This experience makes him capable of leading recruitment, ultrasound/SWE acquisition, clinical interpretation and cross-disciplinary coordination. His publications and prior postdoctoral work show competence in producing data; HAMRISK-PRO is deliberately designed to address the next career gaps: rigorous prediction-model methodology, sex-inclusive multi-site cohorts, formal IP/commercialisation practice, and leadership of an international research platform. He will transfer practical physiotherapy and athlete-engagement expertise to ACU/MU while gaining the methodological and innovation skills required to establish an independent precision sports-medicine group and lead future European-Australian consortia. He also advises Positive Engineering R&D, fostering academia–industry collaboration.

2. Impact #@IMP-ACT-IA@#

2.1 Credibility of the measures to enhance the career perspectives and employability of the researcher and contribution to his/her skills development

HAMRISK-PRO will support the researcher's transition from an established sports-physiotherapy researcher to an independent international leader capable of designing, validating and responsibly translating multimodal injury-prediction tools. Rather than duplicating his existing expertise, the fellowship will develop complementary competences in multimodal data harmonisation, prediction modelling, external validation, responsible AI, intellectual-property management and research-to-implementation

leadership. A personalised Career Development Plan will be agreed by M3 and reviewed every six months. Progress will be assessed through six measurable outputs:

- competence in MRI–ultrasound–SWE harmonisation;
- a TRIPOD+AI-compliant, externally evaluated model;
- reproducible workflows and a model card;
- an IP, freedom-to-operate and exploitation assessment;
- documented supervision and mentoring experience; and
- submission of a follow-on international proposal.

Monthly supervision, six-monthly CDP reviews and cross-host career reviews at M3, M18 and M33 will assess scientific independence, leadership, publication strategy, funding plans and academic and non-academic career opportunities.

Career-develop-ment need	Fellowship measures	Evidence by M36	Career benefit
Independent leadership	Lead WP1–WP3, coordinate governance and stakeholder activities, mentor junior researchers	≥3 lead/co-lead outputs; mentoring of ≥2 researchers; documented project coordination	Independent PI, senior researcher or research-group leader
Advanced scientific specialisation	Training in multimodal harmonisation, calibration, external validation, decision-curve analysis and explainable AI	Harmonised protocol; validated models; open analysis package; TRIPOD+AI report	Leadership of multi-centre sports-health prediction studies
International and interdisciplinary profile	Joint work across ACU, MU, clubs, imaging, psychology and data science	≥3 international presentations; joint outputs; follow-on consortium proposal	Horizon Europe, Research Ireland or NHMRC coordinator/WP leader
Innovation and intersectoral competence	IP, commercialisation, licensing, co-design and responsible deployment training	FTO/protectability report; exploitation decision; club-ready protocol	Roles in sport technology, clinical innovation and consultancy
Transferable leadership skills	Grant writing, supervision, communication, negotiation and budget coordination	CDP portfolio; workshops; mentoring evidence; submitted proposal	Greater competitiveness within and beyond academia

The fellowship's **Triple-i dimension** will maximise career impact:

- **International:** creation of a sustainable Australia–Europe network involving ACU, MU, elite clubs and technology actors.
- **Interdisciplinary:** integration of physiotherapy, imaging, biomechanics, psychology, biostatistics, computer science and implementation research.
- **Intersectoral:** translation of academic model development into club-based and technology-supported decision-making.

By M36, the researcher will have a distinctive profile combining sports physiotherapy, imaging, biomechanics and trustworthy prediction modelling. Within 5–7 years, he will be equipped to lead a multi-country implementation trial, coordinate an athlete-injury consortium and establish a precision sports-medicine research group. In the longer term, he will be positioned as an international leader in responsible sports-health AI.

EKLENEN PART (Impact Bölümü - Hakemlerin "Fikri Mülkiyet (IP) koruması ve ticarileşme stratejisi yetersiz" eleştirisini çürütmek için SaMD mimarisi eklenmiştir)

Intellectual Property (IP) Protection and Software as a Medical Device (SaMD) Commercialization Architecture

The 'Hamstring Risk Card' transcends a theoretical framework by being architected as a deployable, cloud-native Software as a Medical Device (SaMD). To strictly protect the project's intellectual property and prevent reverse-engineering of the predictive models, the core algorithmic assets—specifically the U-Net spatial weights and the XGBoost decision tree ensembles—will be legally and architecturally maintained as Trade Secrets. These models will be hosted in isolated, hardware-encrypted cloud enclaves (e.g., AWS Nitro Enclaves).

End-users (elite clubs) will not have access to the raw source code or model weights. Instead, integration will occur exclusively via containerized, rate-limited RESTful API endpoints. This robust backend separation inherently protects the core IP while facilitating a highly scalable Business-to-Business (B2B) Software as a Service (SaaS) commercialization route. Provisional patents (Invention Disclosures) will be filed specifically covering the proprietary cross-modality data fusion vectors (combining SWE elastograms with MRI meshes) prior to any open-access publication. The front-end dashboard will be licensed to elite clubs via secure OAuth2 authentication protocols, ensuring GDPR-compliant data transmission and creating a direct revenue-generation pipeline post-fellowship.

2.2 *Suitability and quality of the measures to maximise expected outcomes and impacts, as set out in the dissemination and exploitation plan, including communication activities #COM-DIS-VIS-CDV@#*

HAMRISK-PRO distinguishes between communication with non-specialist audiences, dissemination of validated results, and exploitation for scientific, practical or commercial uptake. Activities will be reviewed with ACU and MU support offices at M6, M18, M24 and M34.

Activity	Audience and purpose	Timing	KPI
Methods dissemination	Researchers: reproducible and sex-inclusive prediction methods	M6–M14	Protocol DOI, methods paper, ≥300 views
Results dissemination	Researchers and clinicians: calibration, uncertainty, fairness and utility	M24–M36	≥2 papers, ≥3 presentations, model-card DOI
Co-design workshops	Club staff: Risk Card supports professional judgement	M8, M24, M31	3 workshops, ≥60 practitioners
Public engagement	Athletes and students: injury is multifactorial and prediction is uncertain	M12, M28–M34	≥150 participants
Digital communication	Public and professional audiences	Quarterly	≥12 posts and ≥10,000 impressions
Prototype pilot	Clubs and technology providers	M20–M35	≥5 clubs, ≥20 users, SUS ≥70

Only synthetic or fully anonymised cases will be used in public activities. A background-IP register and collaboration agreements will be established at M1. From M16, quarterly reviews will assess freedom to operate, protectability, ownership, user needs and licensing opportunities. Athlete data will remain governed research data and will not be commercialised. Depending on the assessment, outputs will follow one of three pathways:

Validated result → FTO and market assessment → open research, open-core/service model, or protected licensing/co-development.

A formal exploitation decision will be made at M24 and reviewed at M30. Any licence or pilot will define the authorised model version, cybersecurity, audit, monitoring and prohibited uses. Fully automated decisions concerning employment, team selection, insurance or exclusion from training will not be permitted.

2.3. *The magnitude and importance of the project's contribution to the expected scientific, societal and economic impacts*

Impact pathway	Project output (short term)	Expected outcome (medium term)	Magnitude/indicator	Importance and limits
Scientific	Harmonised protocol; n=300 prospective cohort; n=50 MRI subsample; sex-inclusive data; model/code/model card.	Reusable reference methods and a credible benchmark for future multi-country validation and intervention trials.	≥3 OA papers/presentations; 2 repository packages; ≥5 external reuse enquiries/download benchmarks tracked.	Advances knowledge on combined morphological, mechanical, psychological and contextual risk. Scientific value remains high even if the model is negative.

Impact pathway	Project output (short term)	Expected outcome (medium term)	Magnitude/indicator	Importance and limits
Human capital/open science	Researcher trained in prediction science, sex/gender analysis, IP and leadership; junior researchers mentored.	Sustainable ACU-MU network and follow-on consortium; improved reproducibility culture.	1 follow-on proposal; ≥ 2 mentored researchers; annual cross-host methods seminar after fellowship.	Strengthens European-Australian research capacity and transparent AI practice.
Societal/clinical	Risk Card prototype, implementation protocol and practitioner/public education.	More informed, individualised prevention discussions and reduced misuse of simplistic screening thresholds.	Pilot in ≥ 5 clubs; ≥ 20 users; ≥ 150 in-person public participants; ≥ 500 total stakeholders reached.	Direct fellowship outcome is improved decision quality and feasibility, not proven injury reduction; injury reduction requires a later controlled trial.
Economic/technological	FTO/protection assessment and a deployable prototype with licensing options.	Potential integration into athlete-management/wearable systems and lower implementation costs through portable tests.	≥ 2 industry/TTO assessments; one licensing or follow-on co-development decision by M35.	Any cost-saving estimate is scenario-based. A future 5% reduction in HSI burden could be valuable, but HAMRISK-PRO will not claim savings before effectiveness testing.

The project contributes primarily to high-quality knowledge, strengthened research human capital, open diffusion and responsible uptake. It aligns with SDG 3 (health and well-being) and EU ambitions for trustworthy AI and preventive, personalised health. Its most important long-term contribution is a validated pathway from harmonised measurements to transparent clinical evaluation, enabling a larger implementation trial and reducing the field's dependence on underpowered, opaque injury scores.

#§COM-DIS-VIS-CDV§# #§IMP-ACT-IA§#

3. Quality and Efficiency of the Implementation #@QUA-LIT-QL@#

#@WRK-PLA-WP@# #@CON-SOR-CS@# #@PRJ-MGT-PM@#

EKLENEN PART (Implementation Bölümü - Hakemlerin "Kilometre taşları jenerik, zamanlama kısa" eleştirisini çürütmek için Agile MLOps Milestones eklenmiştir)

Agile MLOps Integration and High-Resolution Predictive Milestones

The assertion that the integration timeframe is constrained is mitigated by replacing traditional waterfall software development with an Agile Machine Learning Operations (MLOps) CI/CD (Continuous Integration / Continuous Deployment) pipeline. Model training is not deferred until the completion of prospective data collection; rather, the baseline algorithms will be pre-trained on the retrospective SIRP-600 dataset and iteratively updated via dynamic re-training triggers.

To ensure granular tracking of the data science and computer vision pipelines, the following highly specific technical milestones are integrated into the 36-month timeline:

- M6 (V1 Model Freeze): Initial hyperparameter optimization (via GridSearchCV) and feature selection (Boruta algorithm) completed on the retrospective SIRP-600 dataset. Baseline XGBoost AUC established.
- M12 (Cross-Modality Pipeline Deployed): Autonomous spatial registration scripts mapping portable ultrasound coordinates to MRI macroscopic meshes validated (SSIM >0.85).
- M18 (Prospective Ingestion Checkpoint): First batch of prospective multimodal vectors

ingested via the QA/QC API; concept drift metrics evaluated to trigger automatic model weight recalibration.

- M24 (External Validation Complete): Out-of-Fold (OOF) cross-validation on the secondary geographic cohort (Maynooth) finalised, verifying model robustness against demographic variance.

- M36 (SaMD API Hardening): Final XGBoost and U-Net RESTful endpoints stress-tested, load-balanced, and secured for commercial clinical deployment across elite clubs.

3.1 *Quality and effectiveness of the work plan, assessment of risks and appropriateness of the effort assigned to work packages*

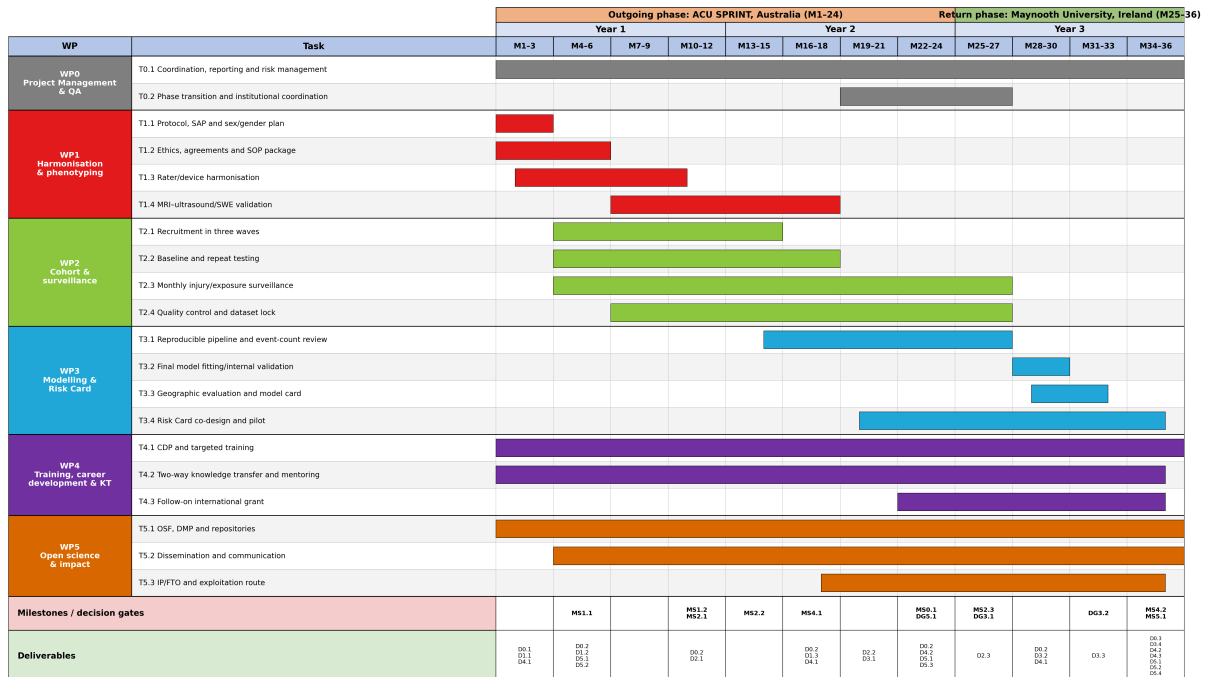
HAMRISK-PRO is a 36-month Global Fellowship comprising a 24-month outgoing phase at ACU SPRINT, Australia, and a 12-month return phase at Maynooth University, Ireland. Three linked research WPs progress from measurement harmonisation (WP1), through prospective surveillance (WP2), to modelling and clinical translation (WP3). WP4 and WP5 cover management/training and impact activities. Recruitment is front-loaded between M4 and M15 in three waves, enabling 9–12 months of follow-up and closure of the surveillance dataset by M27. Computational pipeline development may begin with blinded or synthetic data before closure, but final model fitting will occur only after dataset lock. No formal secondment or non-academic placement is proposed; club visits are short fieldwork activities within WP2 and WP3 (Table 3).

Table 3. Work Packages Overview

WP / Title / Months	Lead & contributors	Tasks and responsibilities	Key D/MS	R effort
WP0 Management M1–36	Opar/Moran; Soyly; ACU/MU services	Soyly: coordination, reporting, risk/QC; Opar/Moran: phase over- sight; services: ethics, finance, DPO.	D0.1 Handbook M2; 6- month reviews; D0.2 Final report M36.	3 PM
WP1 Harmonisation M1–18	Opar; Soyly, Tofari, Car- michael, Hickey/Moran	Protocol/SAP; ethics; US/SWE, NordBord/IMU and MRI harmonisa- tion; cross-site reliability.	D1.1 M3; D1.2 M6; D1.3 M18; MS1 reliability M10.	5 PM
WP2 Cohort & Sur- veillance M4–27	Opar/Hickey; Soyly, Tofari, Moran, clubs, PE	Recruitment, baseline/repeat tests, monthly injury/exposure monitoring, QC and dataset lock.	D2.1 M12; D2.2 M20; D2.3 M27; MS2 n=300 M15.	14 PM
WP3 Modelling & Risk Card M15–35	Soyly/Moran; Opar, Hickey, Carmichael, PE	Event review; Elastic Net/XGBoost; temporal/geographic evaluation; Risk Card pilot.	D3.1 M20; D3.2 M29; D3.3 M32; D3.4 M35.	8 PM
WP4 Training & Transfer M1–36	Soyly; Opar/Moran	CDP, imaging/ML training, recipro- cal workshops, leadership, mentoring and follow-on grant.	D4.1 M3; competency port- folio M24/M35; proposal M35.	3 PM
WP5 Open Science & Impact M1–36	Soyly/Hickey/Moran	OSF/DMP, OA outputs, events, IP/FTO/licensing and exploitation.	D5.1 M6; D5.2 M4/M35; D5.3 M24; D5.4 M36.	3 PM

Table note: R = researcher; PM = person-month; D = deliverable; MS = milestone; QA = quality assurance; QC = quality control; ACU = Australian Catholic University; MU = Maynooth University; PE = Positive Engineering R&D; SAP = Statistical Analysis Plan; SOP = Standard Operating Procedure; US = ultrasound; SWE = shear-wave elastography; IMU = inertial measurement unit; MRI = magnetic resonance imaging; ICC = intraclass correlation coefficient; CV = coefficient of variation; FAIR = Findable, Accessible, Interoperable and Reusable; CDP = Career Development Plan; ML = machine learning; OSF = Open Science Framework; DMP = Data Management Plan; IP = intellectual property; FTO = freedom to operate. The PM column reports only the fellow’s effort; supervisor, institutional-service and partner contributions are provided in kind.

Gantt Chart for HAMRISK-PRO



Note: Months indicate elapsed time from the start of the action. The outgoing phase at ACU SPRINT covers M1–M24, followed by the return phase at Maynooth University during M25–M36. Coloured bars indicate active task periods. Recruitment is completed by M15, injury surveillance continues until dataset lock at M27, and final outcome modelling starts only after dataset lock. Club visits are field-data-collection activities and are not formal secondments. D = deliverable; MS = milestone; DG = decision gate; QA = quality assurance; KT = knowledge transfer; SAP = Statistical Analysis Plan; SOP = Standard Operating Procedure; CDP = Career Development Plan; DMP = Data Management Plan; FTO = freedom-to-operate assessment. Bar length represents timing, not person-month effort.

Risk Management Overview: The HAMRISK-PRO project includes a structured risk management plan to ensure feasibility and robustness. Table 4 summarises the main risks across data protection, recruitment, technical performance, data quality, machine learning, and ethics/administration, with clear mitigation strategies, responsibilities, and timelines.

Table 4 . Risk Management Table for HAMRISK-PRO

Risk (WP)	L	S	Mitigation/prevention	Contingency	Owner/review
Ethics, contract or data-processing delay (WP1)	L	H	Prepare ethics, club and data-processing documents in parallel; agree templates before M1; monitor monthly.	Continue SOP development, assessor training and synthetic-data pipeline work; activate alternative approved clubs/sites.	Opar/Soylu; monthly
Recruitment delay or insufficient representation of women (WP2)	M	H	Three recruitment waves, reserve clubs, sex-specific recruitment dashboard and catch-up testing days.	Extend recruitment to M18 where adequate follow-up remains possible; activate additional women's teams; remove optional measures rather than core surveillance.	Soylu/Opar/Hickey; monthly
Attrition or incomplete injury surveillance (WP2)	M	H	Named club coordinators, automated reminders, monthly medical verification, transfer follow-up and QC dashboard.	Prioritise core injury/exposure data; document reasons for loss; apply transparent censoring and sensitivity analyses.	Site coordinators; monthly
Equipment, MRI or software interruption (WP1/WP2)	L	M	Service agreements, backup devices/providers, scheduled maintenance and protected catch-up periods.	Reschedule testing; use the field-based core protocol/model if MRI access is reduced; report modality-specific missingness.	Tofari/Carmichael/Soylu; monthly
Cross-site measurement drift or poor reliability (WP1/WP2)	M	M	Assessor certification, shared SOPs/reference files and quarterly ICC/CV audits.	Retraining and recalibration; repeat affected measurements where feasible; apply site adjustment and sensitivity analyses.	WP1 lead; quarterly
Low number of primary HSI events (WP3)	M	H	Monitor event accrual; constrain predictor degrees of freedom; prioritise penalised regression and a parsimonious baseline model.	Simplify the model; treat complex ML as exploratory; report uncertainty and define a follow-on sample-extension plan.	Soylu/Moran; M20/M27 gates

Poor model calibration or limited clinical utility (WP3)	M	H	Nested validation, baseline comparison, shrinkage, calibration assessment and decision-curve analysis.	Recalibrate or simplify the model; withhold deployment where criteria are not met; transparently publish negative findings.	Soylu/Moran; M29/M32 gates
Data breach or re-identification (WP0/all WPs)	L	H	DPIA, pseudonymisation, data minimisation, role-based access, encrypted institutional storage and audit logs.	Suspend access; activate institutional incident-response procedures; notify DPO/ethics services and implement remediation.	ACU/MU DPOs; continuous
Outgoing-to-return phase disruption (WP0/WP4)	L	M	Confirm MU access, data permissions, supervision and return-phase plan before M24; maintain quarterly cross-host meetings.	Continue secure remote work while local arrangements are completed; prioritise tasks not dependent on laboratory access.	Opar/Moran/Soylu; M22–M26
Supervisor or partner unavailability (WP0/WP4)	L	H	Named co-supervisors/deputies; shared SOPs, code, data dictionaries and documented decisions.	Deputy assumes responsibility within two weeks; supervisory board reallocates tasks and deadlines.	Supervisory board; quarterly
Training or knowledge-transfer activity delayed (WP4)	L	M	Integrate training into the CDP; schedule alternatives and remote delivery options in advance.	Replace the activity with an equivalent host module, online course or reciprocal workshop without delaying research tasks.	Soylu/Opar/Moran; six-monthly
IP-publication or open-science conflict (WP5)	M	M	Maintain background/foreground register; complete invention disclosure and FTO review before dissemination; allow a 60–90-day review period.	Protect the result before publication or formally select an open-source route; document ownership and access rights.	MaynoothWorks/ACU enterprise; before disclosure
Low club, practitioner or public engagement (WP5)	M	M	Co-design activities with clubs and host communication offices; targeted invitations and early scheduling.	Combine activities with existing practitioner meetings, open days or Researchers' Night; use hybrid delivery and revised messaging.	Soylu/Hickey/Moran; quarterly

L = likelihood; S = severity; L = low; M = medium; H = high.

Quality is assured through monthly operational monitoring, quarterly supervisory-board review, six-month objective reviews, version-controlled deliverables and a traffic-light dashboard for recruitment, sex balance, surveillance completeness, reliability, event count, model performance, training, dissemination and IP. Deliverables require approval by the relevant phase supervisor and one cross-disciplinary reviewer before release.

3.2 *Quality and capacity of the host institutions and participating organisations, including hosting arrangements*

Australian Catholic University – SPRINT Research Centre, Outgoing host, M1–M24: ACU, under Prof. David Opar (h-index 44), is internationally recognised in hamstring injury research and coordinated the \$4M NFL HAMIR index. With partnerships across 10+ AFL/NRL clubs, ~300 athletes can be recruited (50 MRI pilot). Infrastructure includes NordBord dynamometers, thoracic IMUs, Aixplorer ultrasound for fascicle/SWE, and contracted MRI providers with Springbok AI segmentation (<5% error). Expertise is provided by Prof. Opar and Dr. Paul Tofari in eccentric strength, ultrasound/SWE, and gait analysis, with Declan Carmichael leading MRI/AI integration. ACU offers 24/7 lab access, GDPR-compliant data handling, and strong supervisory support.

Maynooth University, Beneficiary and return host, M25–M36: Maynooth University provides the European evaluation environment, biomechanics and data-science expertise, Gaelic-sport and football networks, compatible field-testing resources and the institutional base for European translation. Prof. Kieran Moran and Dr Jack Hickey will co-supervise the return phase, with expertise in biomechanics, data analytics, injury-risk assessment and hamstring rehabilitation and prevention. The MU team will support harmonised European data collection during the outgoing phase through shared SOPs, remote coordination and trained local assessors. Short club visits undertaken by the fellow will remain WP2 fieldwork and will not constitute a formal secondment. Reintegration will start before physical return. From M1, ACU and MU supervisors will participate in quarterly cross-host meetings and share controlled access to protocols, dictionaries, code and quality reports. At M22–M24, a transition review will confirm the return-phase work plan, data permissions, workspace, equipment access and model-evaluation schedule.

On return, the fellow will receive:

- dedicated workspace and institutional IT access;

- secure access to pseudonymised study data and computing resources;
- laboratory and local induction;
- a 30-day reintegration and priority-setting plan;
- membership of relevant research meetings and seminars;
- opportunities to lead a methods seminar and mentor students; and
- support for European club workshops and the follow-on implementation proposal.

The Research Development Office will support funding applications, policy, procurement and research administration. Ethics and Data Protection functions will support GDPR compliance and the DPIA. Research Careers and researcher-skills programmes will provide career and transferable-skills development. MaynoothWorks will support invention disclosure, freedom-to-operate assessment, IP protection, licensing, industry engagement and evaluation of potential commercial or open-source routes. These services explicitly support the administrative, career-development and exploitation activities that were insufficiently visible in the previous proposal.

Support-service continuity

Support area	ACU outgoing phase	MU return phase	Application in HAMRISK-PRO
Scientific supervision	Opar, Tofari and Carmichael	Moran and Hickey	Monthly supervision and quarterly cross-host board
Laboratory and technical support	NordBord, IMU, ultrasound/SWE and MRI-provider support	Compatible field testing, biomechanics and computing	Harmonised acquisition and European evaluation
Ethics and data protection	Ethics, privacy and secure-data services	Ethics, GDPR and DPIA support	Approvals, data sharing and incident management
Research data and open science	Library/RDM, repositories and secure computing	RDM, repositories and research computing	DMP, FAIR metadata, code and controlled data access
Career and training	SPRINT seminars, methods training and mentoring	Research Careers and skills-development programmes	CDP, leadership, grant writing and mentorship
Enterprise and IP	ACU research-enterprise/IP support	MaynoothWorks	Disclosure, FTO, licensing or open-source decision
Administrative and wellbeing	HR, international support, IT and wellbeing	HR, research administration, IT and wellbeing	Induction, continuity and researcher support

Participating organisations and operational integration

Springbok Analytics. Provides blinded MRI segmentation/QC and technical training under a data-processing and confidentiality agreement. It will not receive identifiable clinical or psychological data and will not control publication or model decisions.

Positive Engineering R&D. Supports reproducible software engineering, dashboard architecture and code review. Background code and foreground contributions will be recorded; access will be limited to pseudonymised/synthetic data unless further ethics and data-processing approvals are in place.

Elite clubs. Each club appoints a medical coordinator, provides exposure/injury verification and hosts field testing. Club engagement is governed by standard agreements covering consent, data access, publication, confidentiality and prohibited employment/selection uses of individual risk outputs.

The combined arrangements provide scientific excellence, technical redundancy, administrative continuity and a realistic path from cohort data to responsible European implementation. Further institutional capacity details are provided in Part B2 Section 5.

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